



ARDUINO-BASED HUMAN DETECTION PROBE

Technical Report



25 APRIL 2016
INTERNET OF THINGS 261
Tshiamo Mosweu

1. Introduction

This report details the design and implementation of a Human Detection Probe using an Arduino Uno. After an earthquake, the system uses a PIR sensor to find people behind rubble, sounds and sees sirens, and shows status updates on an LCD. The project integrates sensors, actuators, and real-time input to illustrate IoT concepts.

2. Methodology

Components

- Arduino Uno
- PIR Sensor
- Piezo Buzzer
- Red LED
- 16x2 I2C LCD
- PushButton
- Resistors

Circuit Design

The circuit, designed in TINKERCAD, includes the following key connections:

- PIR Sensor
 - Output to Arduino D2
 - VCC to 5V
 - GND to GND
- LED: Digital pin D7 with a 220ohm resistor
- Buzzer: Digital pin D8
- Pushbutton: Digital pin D3 with a 10kohm pull-down resistor
- I2C LCD:
 - SDA to Arduino A4
 - SCL to Arduino A5

Arduino Sketch

The code performs the following tasks:

1. Initializes the I2C LCD and configures I/O pins.
2. Reads the PIR sensor.
3. Triggers alarms upon human detection.
4. Resets the alarm via the push button.
5. Updates the LCD with real-time status.

3. Results

Functional Testing

- Human Detection: The PIR sensor reliably triggered alarms when motion was detected.
- Alarm Reset: The pushbutton successfully silenced the buzzer and reset the LED/LCD.
- LCD Display: Updated dynamically between "Status: Normal" and "Human Detected!".

Challenges and Solutions

- PIR Sensor Stability: Added a 5-second delay after reset to avoid false triggers
- I2C LCD Address Conflict: Adjusted the I2C address (0x27) to match the hardware.

4. Discussion

Design Improvements

- Pull-Up Resistor for PIR: The PIR sensor's output pin (D2) lacks a pull-up resistor. Adding a 10kΩ resistor or enabling INPUT_PULLUP in code would improve stability.
- Debounce Button: Implement software debouncing to prevent accidental resets.
- Standard LCD: The project specifies a 16x2 LCD, but the I2C variant was used to save pins. Include both options in the report.

Market Readiness

- Enclosure Design: Waterproof casing for debris environments.
- Battery Backup: Integrate a 9V battery for portability.

5. Conclusion

The Human Detection Probe effectively demonstrates human detection using IoT components. While functional, minor hardware adjustments (e.g., pull-up resistor) and code optimizations (debouncing) would enhance reliability.